Task 02: Judge the Type of Operating System for Each Environment

Operating Systems (CSA04) — Assignment 1 (CO1)

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Marks: 3 Marks (Bloom's: K5 - Evaluate)
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> ***Task 2 Description:*** *Compare OS types (mainframe, desktop, multiprocessor, distributed, clustered, real-time). Select the best fit for each hospital environment with rejected alternatives. Contrast SMP vs. AMP for the imaging server, and evaluate database clustering with fail-over vs. a single large multiprocessor machine.*

1. SYSTEM TYPE SELECTION & REJECTION MATRIX

Environment	Chosen OS Type	Rejected Alternatives	Defended Architectural Rationale
Bedside Monitors	**Hard Real-Time OS** (e.g., QNX Neutrino, FreeRTOS)	Desktop OS, Mobile OS, Soft RTOS	Guarantees deterministic interrupt handling and dispatch latency (< 100µs). Eliminates priority inversion via priority inheritance protocols.
Nurses' Tablets	**Handheld Mobile OS** (e.g., Android AOSP)	Desktop OS (Ubuntu/Win), Bare RTOS	Built-in touch stack, aggressive wake-lock duty-cycling for 12-hour shifts, and per-app sandbox memory protection.
Doctors' Workstations	**Desktop Interactive OS** (e.g., Enterprise Linux / Win11)	Batch OS, Embedded RTOS	Supports high-resolution multi-monitor DICOM viewers, smart-card hardware authentication, and interactive multitasking.
Central Database	**Clustered OS with Fail-Over** (e.g., RHEL HA / Pacemaker)	Single SMP Mainframe, Desktop OS	Active-passive or active-active quorum nodes guarantee sub-5s automatic fail-over, meeting the 24/7/365 zero-downtime mandate.
Imaging Server	**Multiprocessor SMP OS** (e.g., Linux SMP + OpenMP/CUDA)	Asymmetric Multiprocessor (AMP), Distributed Grid	High-bandwidth shared physical memory enables parallel 3D Fast Fourier Transforms without network serialization overhead.
Partner Hospital Link	**Distributed Network OS** (e.g., Kubernetes Gateway / Linux)	Clustered Local OS	Decoupled microservice proxies handle WAN latency, heterogeneous remote procedure calls (RPC), and TLS 1.3 token exchange.

2. MULTIPROCESSING ARCHITECTURE FOR MRI IMAGING: SMP VS. AMP

Symmetric Multiprocessing (SMP) — Chosen: All CPU cores share a single operating system instance, a unified physical memory address space, and an integrated thread scheduler. Threads performing parallel 3D slice reconstructions access shared image matrices directly via pointers without inter-core data copying.

Asymmetric Multiprocessing (AMP) — Rejected: Cores are statically partitioned with dedicated OS instances or static tasks (e.g., Core 0 runs supervisor tasks, Cores 1–7 run fixed reconstruction steps). AMP introduces significant IPC messaging overhead, complex core-to-core memory copying, and static load imbalances when scan sizes vary.

3. ALTERNATIVE CONCEPT: SINGLE LARGE MULTIPROCESSOR VS. CLUSTERED FAIL-OVER

- **Availability & Single Point of Failure (SPOF):** A single large multiprocessor machine (e.g., 64-socket SMP) presents a critical SPOF. Any motherboard fault, power supply failure, or kernel panic halts the entire hospital EHR. In contrast, an HA cluster with shared-nothing nodes and heartbeats triggers automatic fail-over within 3 to 5 seconds.
- **Throughput & Scaling (Time & Space Complexity):** As processor counts increase on a single SMP machine, shared bus and NUMA interconnect contention limit performance scaling: `Throughput = O(N / (1 + α(N-1)))` (Universal Scalability Law). In contrast, clustered nodes scale horizontally at significantly lower hardware cost: $Cost_{SMP} >> 3 \times Cost_{Cluster_Node}$.